

## NAANMUDHALVAN

Project Report

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# Implement Client Script & UI Policy (Incident)

*Enforcing data integrity on ServiceNow Incident records*

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|              |                                                       |
|--------------|-------------------------------------------------------|
| Team ID      | SWTID-2026-4375                                       |
| Team Size    | 5                                                     |
| Team Leader  | SUPRATHIC K G S                                       |
| Team Members | Sudharsan S<br>Thatshana B<br>Thiyaku R<br>Surendar M |
| Platform     | ServiceNow                                            |
| Program      | Naan Mudhalvan, through SmartBridge (MySkillWallet)   |

Project documentation prepared by Thatshana B

# 1. INTRODUCTION

## 1.1 Project Overview

Incident records are where most service desk work begins. They are triaged, routed to the right group and measured against SLAs, so the quality of the information entered when an incident is logged has a direct effect on how quickly and how correctly it gets resolved. When that quality depends only on people remembering to fill in the right things, gaps and inconsistencies are bound to appear.

This project, Implement Client Script & UI Policy (Incident), shows how ServiceNow's client-side controls can protect data quality at the moment of entry. Rather than trusting manual checks, we configured the Incident form so that it responds to the impact of the incident, fills in values where that makes sense, and refuses to save a record that breaks a business rule.

The solution is made up of six connected pieces of work:

- **UI Policy "High Impact Control"**: applies to the Incident table when Impact is 1 - High, makes Assignment group mandatory, and reverses itself automatically when the condition no longer holds.
- **UI Policy Action for Urgency**: makes the Urgency field read-only while the policy condition is met.
- **onChange Client Script**: sets Urgency to High as soon as Impact is changed to High.
- **onSubmit Client Script**: stops a High impact incident from being saved when Assigned to is empty.
- **onCellEdit Client Script**: prevents the State field from being changed directly in the list view.
- **Structured testing**: covering mandatory enforcement, successful save, the reverse condition, list-edit blocking and form-based updates.

Skills covered: Incident Management, UI Policy, UI Policy Actions, Client Scripts and Form Validation.

## 1.2 Purpose

The purpose of this project is to demonstrate, through a working configuration, how data integrity can be enforced directly on the Incident form. Specifically, the project aims to:

- Make fields mandatory only when a defined condition is true, and return them to normal when it is not.
- Auto-populate values so that related fields, such as Impact and Urgency, stay consistent.
- Control field behaviour, for example by locking a field as read-only.
- Prevent record submission when required conditions are not met, and tell the user why.
- Improve the quality and consistency of Incident data, which in turn supports reporting accuracy, SLA compliance and overall service quality.

Because the scope is deliberately small and focused, the project also works as a compact, practical introduction to the way UI Policies and Client Scripts complement each other.

# 2. IDEATION PHASE

## 2.1 Problem Statement

Incident records need consistent and accurate data to be triaged, routed and resolved effectively. In practice, relying only on user awareness and manual checks means that incomplete, inconsistent or incorrect data can still be submitted. This affects reporting accuracy, SLA compliance and the overall quality of service. What is needed is a way to enforce conditional field behaviour and validation at the user interface level, so that an Incident meets the defined business rules before it is submitted.

| Aspect          | Description                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------|
| Who is affected | Users who create and update Incident records, and the teams who depend on that data for triage, routing and reporting. |

| Aspect                       | Description                                                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>What is needed</b>        | Incident records that are complete, consistent and compliant with business rules at the time they are saved.                    |
| <b>The problem</b>           | Manual checks and user awareness alone allow incomplete, inconsistent or incorrect data to be submitted.                        |
| <b>Impact of the problem</b> | Reduced reporting accuracy, weaker SLA compliance and lower service quality.                                                    |
| <b>Approach taken</b>        | Use UI Policies and Client Scripts to control field visibility, mandatory status and save-time validation on the Incident form. |

## 2.2 Empathy Map Canvas

The empathy map below looks at the situation through the eyes of a person who logs or updates incidents. It is drawn from the problem described in the project brief rather than from a formal survey, and it helped us decide which behaviours the form should guide or restrict.

|                                                                                                                                       |                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SAYS</b><br>"Which fields do I have to fill in for a high-impact incident?"<br>"Why won't this record save?"                       | <b>THINKS</b><br>The incident is filled in well enough for now.<br>Details such as assignment can be sorted out later.                         |
| <b>DOES</b><br>Logs incidents quickly and may leave key fields empty.<br>Changes fields directly, including State from the list view. | <b>FEELS</b><br>Under time pressure and unsure which rules apply.<br>Frustrated when a problem is discovered late or without explanation.      |
| <b>PAINS</b><br>Incomplete, inconsistent or incorrect submissions.<br>Rework, SLA risk and unreliable reports.                        | <b>GAINS</b><br>Clear, immediate guidance on the form itself.<br>Fields that fill in or lock automatically, and problems caught before saving. |

## 2.3 Brainstorming

We considered several ways to keep Incident data clean and weighed each against the problem statement. The table records the ideas and what we decided.

| # | Idea                                                                | What it would achieve                                                                | Decision   |
|---|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------|
| 1 | Rely on user awareness and manual checks                            | Nothing new; this is the situation the brief identifies as the cause of the problem. | Not chosen |
| 2 | UI Policy that makes Assignment group mandatory when Impact is High | Conditional mandatory field without any code.                                        | Selected   |
| 3 | UI Policy Action that makes Urgency read-only                       | Stops users overriding Urgency on high-impact incidents.                             | Selected   |
| 4 | onChange script that sets Urgency when Impact becomes High          | Automatic population so Impact and Urgency stay aligned.                             | Selected   |
| 5 | onSubmit script that blocks saving without Assigned to              | Save-time validation with a clear error message.                                     | Selected   |
| 6 | onCellEdit script that blocks State changes in the list             | Protects State from quick edits that bypass the form.                                | Selected   |
| 7 | Enable "Reverse if false" on the UI Policy                          | Restores normal behaviour automatically when Impact is no longer High.               | Selected   |

### 3. REQUIREMENT ANALYSIS

#### 3.1 Customer Journey Map

The journey below follows a user from opening a new Incident to updating it later, and shows what the system does at each stage.

| Stage                   | What the user does                                       | What the system does                                                                                                               | Result                     |
|-------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1. Start                | Opens Incident → Create New.                             | Loads a new Incident form and evaluates the UI Policy on load.                                                                     | Form is ready.             |
| 2. Set impact           | Selects Impact as 1 - High.                              | UI Policy applies: Assignment group becomes mandatory and Urgency becomes read-only. The onChange script sets Urgency to 1 - High. | Form guides the user.      |
| 3. Complete the form    | Fills in Caller, Short description and Assignment group. | Mandatory fields are marked on the form.                                                                                           | Required data is captured. |
| 4. Submit without owner | Clicks Submit with Assigned to empty.                    | onSubmit script shows an error under Assigned to and cancels the save.                                                             | Record is not saved.       |
| 5. Correct and save     | Selects a user in Assigned to and submits again.         | Validation passes.                                                                                                                 | Record is saved.           |
| 6. Lower the impact     | Changes Impact from High to Medium.                      | Reverse if false undoes the policy actions; Urgency becomes editable and Assigned to is no longer enforced.                        | Record saves normally.     |
| 7. Edit from the list   | Double-clicks State in the incident list.                | onCellEdit script shows an alert and cancels the edit.                                                                             | State is unchanged.        |
| 8. Edit from the form   | Opens the incident and changes State on the form.        | Change is allowed.                                                                                                                 | State is updated.          |

#### 3.2 Solution Requirements

##### Functional requirements

| ID  | Requirement                                                                                                                                     |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|
| FR1 | Create an active UI Policy named High Impact Control on the Incident table, triggered when Impact is 1 - High.                                  |
| FR2 | Make Assignment group mandatory while the condition is met, and reverse this automatically when it is not (Reverse if false).                   |
| FR3 | Make Urgency read-only while the condition is met, leaving its visibility unchanged.                                                            |
| FR4 | When Impact is changed to High, set Urgency to 1 - High automatically. The script must not run during form load or when the new value is empty. |
| FR5 | Block saving when Impact is High and Assigned to is empty, and show an error message against the Assigned to field.                             |
| FR6 | Block edits to the State field made directly in the list view, show an alert and keep the original value.                                       |
| FR7 | Continue to allow State to be changed from the Incident form.                                                                                   |

##### Non-functional and environment requirements

- A ServiceNow instance and a user with administrative or configuration access.
- Lightweight and efficient: each rule is a small configuration or a short script.
- Immediate feedback: users see the result of a rule on the form itself, without waiting for a server response or a later review.
- Reversible: rules that depend on a condition must switch off cleanly when the condition changes.

- Easy to implement and maintain, making it suitable for a short-duration micro project.

### 3.3 Data Flow Diagram

The diagram shows how user actions on the Incident form and list flow into the different client-side controls, and what each control returns. The UI Policy and the onChange script react as the user changes Impact, the onSubmit script runs when the user tries to save, and the onCellEdit script runs when a value is edited inline in a list.

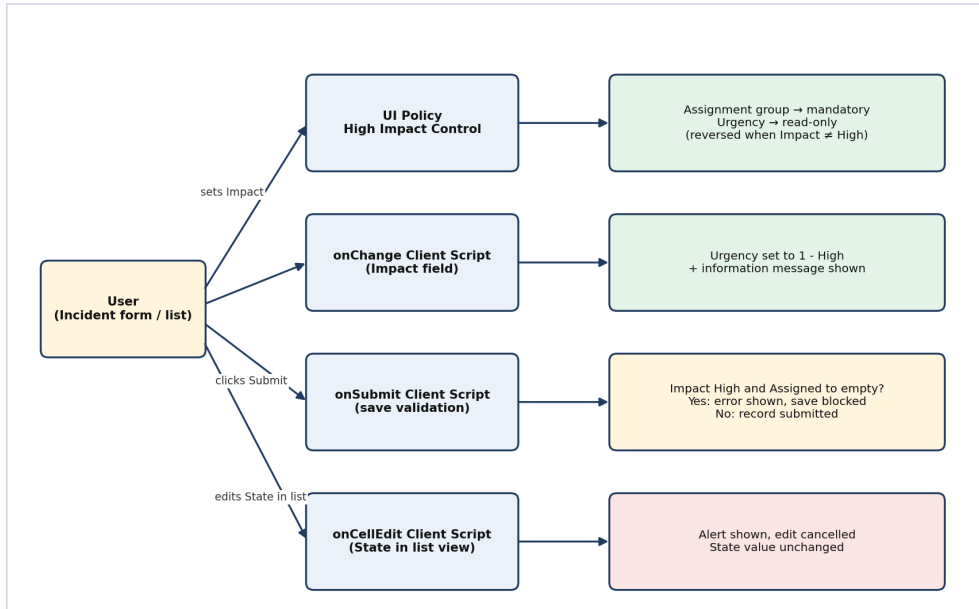


Figure 1: Data flow between user actions, client-side controls and their outcomes.

### 3.4 Technology Stack

| Area                       | Technology / Component                                                                                         |
|----------------------------|----------------------------------------------------------------------------------------------------------------|
| Platform                   | ServiceNow (developer instance used for configuration and testing)                                             |
| Application module         | Incident Management, using the Incident [incident] table                                                       |
| Declarative controls       | UI Policy and UI Policy Actions (System UI → UI Policies)                                                      |
| Scripted controls          | Client Scripts of type onChange, onSubmit and onCellEdit (UI Type: Desktop)                                    |
| Scripting language and API | JavaScript with the GlideForm API (g_form): setValue, getValue, addInfoMessage / showFieldMsg and showErrorBox |
| Access level               | Administrator or configuration access                                                                          |

## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

Each part of the problem has a matching control in the solution, and each control is placed where the user actually works, on the form or in the list.

| Problem observed                                                 | Solution component                                                                   |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| High impact incidents are submitted without routing information. | UI Policy makes Assignment group mandatory; onSubmit script requires Assigned to.    |
| Urgency does not match a high Impact value.                      | onChange script sets Urgency to High; UI Policy Action locks the field as read-only. |
| Incorrect or incomplete records reach the database.              | onSubmit script cancels the save and displays an error message.                      |

| Problem observed                                                   | Solution component                                                          |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------|
| State is changed quickly from the list without opening the record. | onCellEdit script blocks the change and asks the user to open the Incident. |
| Rules should not stay in force after the condition changes.        | "Reverse if false" restores the fields when Impact is no longer High.       |

## 4.2 Proposed Solution

The proposed solution combines one UI Policy with two actions and three Client Scripts. The UI Policy handles behaviour that can be described with a simple condition. The Client Scripts handle the cases that need logic: setting a value, validating at save time, and intercepting a list edit.

| Component                           | Type / trigger                                          | Behaviour                                                                                  |
|-------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------|
| High Impact Control                 | UI Policy, Incident table, condition Impact is 1 - High | Applies its actions while Impact is High; Reverse if false is enabled.                     |
| Action: Assignment group            | UI Policy Action                                        | Mandatory: True. Read only: False.                                                         |
| Action: Urgency                     | UI Policy Action                                        | Read only: True. Visible and Mandatory left alone.                                         |
| Auto set urgency for high impact    | Client Script, onChange on Impact                       | Sets Urgency to 1 when Impact becomes 1 and informs the user.                              |
| Prevent save if assigned to missing | Client Script, onSubmit                                 | Returns false and shows an error on Assigned to when Impact is 1 and Assigned to is empty. |
| Prevent state change via list edit  | Client Script, onCellEdit on State                      | Shows an alert and cancels the list edit.                                                  |

Key configuration details of the UI Policy are summarised below.

| Setting                  | Value               |
|--------------------------|---------------------|
| Short description (name) | High Impact Control |
| Table                    | Incident [incident] |
| Application              | Global              |
| Active                   | True                |
| Order                    | 100                 |
| Condition                | Impact is 1 - High  |
| Global / On load         | Selected / Selected |
| Reverse if false         | Selected            |
| Inherit                  | Not selected        |

## 4.3 Solution Architecture

The architecture has three layers. Users interact with the Incident form and list in the browser. Between them and the database sits the client-side control layer, made up of the UI Policy and the three Client Scripts. Only data that passes these checks is submitted to the Incident table.

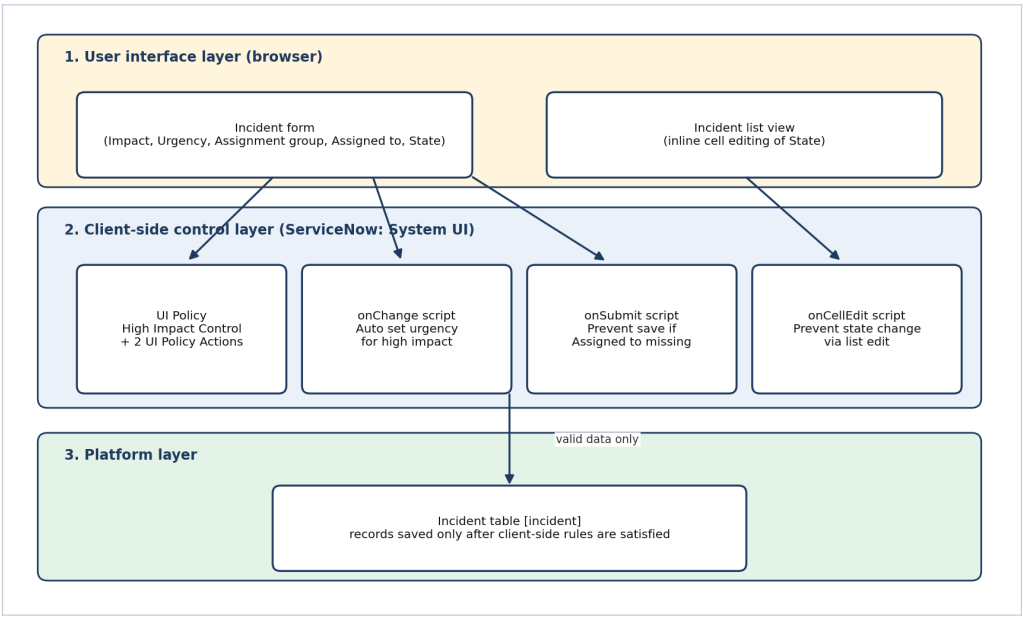


Figure 2: Layered solution architecture for the Incident client-side controls.

## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

The project was organised as six tasks, each building on the one before it. Since it is a micro project, the work was kept short and focused, with configuration first and testing at the end.

| Task   | Activity                         | Work carried out                                                                                                                               | Status    |
|--------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Task 1 | Create UI Policy on Incident     | Created High Impact Control with the condition Impact is 1 - High, added the Assignment group action (mandatory) and enabled Reverse if false. | Completed |
| Task 2 | Create UI Policy Action: Urgency | Added an action on the policy to make Urgency read-only, leaving Visible unchanged.                                                            | Completed |
| Task 3 | Create onChange Client Script    | Wrote and activated "Auto set urgency for high impact" on the Impact field.                                                                    | Completed |
| Task 4 | Create onSubmit Client Script    | Wrote and activated "Prevent save if assigned to missing" for save-time validation.                                                            | Completed |
| Task 5 | Create onCellEdit Client Script  | Wrote and activated "Prevent state change via list edit" on the State field.                                                                   | Completed |
| Task 6 | Test the configuration           | Ran five test activities covering mandatory enforcement, successful save, reverse condition, list-edit blocking and form-based update.         | Completed |

## 6. FUNCTIONAL AND PERFORMANCE TESTING

Testing followed the five activities defined for the project. Each test was run on the ServiceNow instance and the observed result was compared with the expected result.

| Test  | Scenario and steps                                                                                          | Expected result                                                 | Observed result                                                                                                                                                                                    | Status |
|-------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| TC-01 | Mandatory enforcement. Create a new Incident, set Impact to High, leave Assigned to empty and click Submit. | Record is not saved; an error message appears for Assigned to.  | Save was blocked and the message "Assigned To is mandatory for High impact incidents." appeared under Assigned to. Urgency showed 1 - High as read-only and Assignment group was marked mandatory. | Pass   |
| TC-02 | Successful save. Select a user in Assigned to and click Submit.                                             | Record is saved without errors.                                 | With Assigned to filled in (Beth Anglin), no error was raised and the incident INC0010012 was created and appeared in the Incident list.                                                           | Pass   |
| TC-03 | Reverse condition. Open an incident with High impact and change Impact to Medium.                           | Assigned to no longer enforced; Urgency editable; record saves. | With Impact at 2 - Medium, Assigned to could be left empty and Urgency was editable again.                                                                                                         | Pass   |
| TC-04 | List edit blocking. In the Incident list, double-click the State field of a record.                         | Alert appears and State is unchanged.                           | An alert reading "State cannot be changed from the list. Please open the Incident." appeared and the edit was cancelled.                                                                           | Pass   |
| TC-05 | Form-based update. Open the incident, change State and click Update.                                        | State change is saved.                                          | INC0008111 was updated to In Progress from the form and the list showed the new State.                                                                                                             | Pass   |

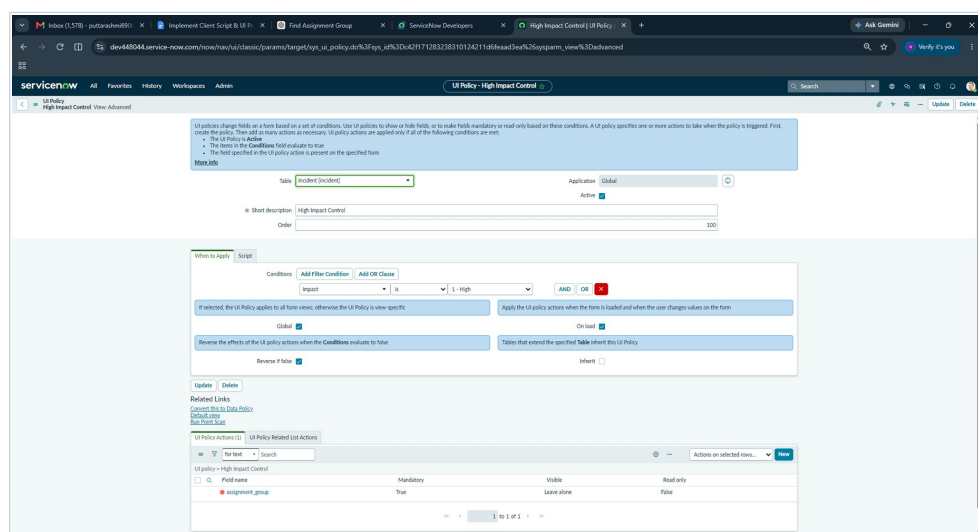
## 6.1 Performance Testing

This project focused on functional correctness, so no formal load or response-time measurements were taken. From a design point of view, the controls are light on resources: the UI Policy is a simple declarative rule, and each Client Script is only a few lines long and runs in the user's browser as the form or list is used. In all the test runs above, the form and list responded immediately to each action.

## 7. RESULTS

### 7.1 Output Screenshots

The screenshots below record the implementation and testing, from the UI Policy configuration to confirming the behaviour on the form and in the list.



**Figure 3:** High Impact Control UI Policy on the Incident table with the condition Impact is 1 - High, Reverse if false enabled, and the Assignment group action (Mandatory: True).

UI Policy Action configuration for Urgency:

- UI policy:** High Impact Control
- Table:** Incident [incident]
- \* Field name:** Urgency
- Application:** Global
- Mandatory:** Leave alone
- Visible:** Leave alone
- Read only:** True
- Clear the field value:** ☐

Buttons: Update, Delete

Figure 4: UI Policy Action for Urgency with Read only set to True.

Client Script configuration for "Auto set urgency for high impact":

- Name:** Auto set urgency for high impact
- Table:** Incident [incident]
- UI Type:** Desktop
- Type:** onChange
- Field name:** Impact
- Script:**

```

function onChange(control, oldValue, newValue, isLoading) {
    if (isLoading || newValue === '') {
        return;
    }
    if (newValue == "1") {
        $form.setFieldValue("urgency", "High");
        $form.setFieldValue("urgency", "set to high for high impact incidents.");
    }
}

```
- Update Versions:**

| Name                                           | Recorded at         | State   | Source             | Reverted from |
|------------------------------------------------|---------------------|---------|--------------------|---------------|
| inc_script_client_4a8f93f4b36330124211d8aad74a | 2024-09-27 23:50:39 | Current | Update Set Default | (empty)       |

Buttons: Update, Delete, Run Default Script

Figure 5: onChange Client Script "Auto set urgency for high impact" on the Impact field.

Client Script configuration for "Prevent save if Assigned to missing":

- Name:** Prevent save if Assigned to missing
- Table:** Incident [incident]
- UI Type:** Desktop
- Type:** onSubmit
- Script:**

```

function onSubmit() {
    if ($form.getFieldValue("Impact") == "1" &&
        $form.getFieldValue("assigned_to") == "") {
        $form.showErrorBox(
            "assigned to",
            "assigned to is mandatory for High impact incidents."
        );
        return false;
    }
    return true;
}

```

Buttons: Update, Delete, Run Default Script

Figure 6: onSubmit Client Script "Prevent save if assigned to missing".

Figure 7: onCellEdit Client Script "Prevent state change via list edit" on the State field.

Figure 8: Impact set to High: Urgency is read-only, Assignment group is mandatory and the save is blocked with an error under Assigned to.

Figure 9: Assigned to filled in with a user, so the incident can be submitted successfully.

Figure 10: Reverse condition test: Impact changed to 2 - Medium, Urgency editable and Assigned to no longer enforced.

| Number     | Opened              | Short description                                               | User                   | Priority     | State       | Category       | Assignment group        | Assigned to   | Updated             | Updated by |
|------------|---------------------|-----------------------------------------------------------------|------------------------|--------------|-------------|----------------|-------------------------|---------------|---------------------|------------|
| INC0010002 | 2024-09-28 03:11:25 | Testing UI policy                                               | Adela Cervantiz        | 2 - High     | In Progress | Inquiry / Help | Software                | (empty)       | 2024-09-28 03:27:30 | admin      |
| INC0009009 | 2018-08-30 01:06:16 | Unable to access the shared folder.                             | David Miller           | 1 - Critical | New         | Inquiry / Help | Application Development | system        | 2024-09-28 02:03:22 | system     |
| INC0009005 | 2018-08-31 21:35:21 | Email server is down.                                           | David Miller           | 1 - Critical | Resolved    | (empty)        | (empty)                 | (empty)       | 2018-12-12 23:18:55 | admin      |
| INC0009001 | 2018-09-11 20:56:26 | Unable to post content on a Wiki page                           | David Miller           | 3 - Moderate | (empty)     | (empty)        | (empty)                 | (empty)       | 2018-12-12 23:32:42 | admin      |
| INC0008112 | 2019-07-29 11:48:43 | Assessment - ATF Assessor                                       | survey user            | 5 - Planning | New         | Inquiry / Help | (empty)                 | (empty)       | 2019-07-29 11:49:28 | admin      |
| INC0008111 | 2019-07-22 14:04:57 | ATF: Test1                                                      | System Administrator   | 5 - Planning | New         | Inquiry / Help | (empty)                 | (empty)       | 2019-07-22 14:05:48 | admin      |
| INC0008001 | 2021-01-15 13:04:14 | ATF: TEST2                                                      | survey user            | 5 - Planning | New         | Inquiry / Help | (empty)                 | (empty)       | 2021-01-21 15:31:42 | admin      |
| INC0007002 | 2018-10-16 22:47:51 | Need access to the common drive.                                | David Miller           | 4 - Low      | New         | Inquiry / Help | (empty)                 | (empty)       | 2018-12-12 23:28:49 | admin      |
| INC0007001 | 2018-10-16 22:47:51 | Employee payroll application server is down.                    | David Miller           | 1 - Critical | New         | Hardware       | Openspace               | (empty)       | 2024-06-12 15:15:11 | system     |
| INC0001990 | 2020-06-07 09:02:25 | Unable to access the personal details section in payroll portal | Problem CoordinatorATF | 5 - Planning | On Hold     | Inquiry / Help | (empty)                 | (empty)       | 2020-06-07 09:04:34 | admin      |
| INC0000599 | 2016-08-10 09:14:29 | Unable to access team file share                                | Rick Bertie            | 3 - Moderate | New         | Inquiry / Help | (empty)                 | (empty)       | 2016-08-10 09:14:29 | admin      |
| INC0000588 | 2016-08-10 09:37:45 | Performance problems with email                                 | Bow Ruggeri            | 5 - Planning | New         | Inquiry / Help | (empty)                 | (empty)       | 2016-08-10 09:37:45 | admin      |
| INC0000557 | 2016-08-10 09:14:59 | Performance problems with wifi                                  | Bertie Luby            | 5 - Planning | New         | Inquiry / Help | (empty)                 | (empty)       | 2016-08-10 09:14:59 | admin      |
| INC0000555 | 2024-05-21 21:47:23 | SAP Sales app is not accessible                                 | Carol Coughlin         | 1 - Critical | In Progress | (empty)        | Service Desk            | Beth Anglin   | 2024-06-11 12:40:45 | admin      |
| INC0000554 | 2015-11-02 12:49:08 | SAP Materials Management is slow or there is an outage          | Christen Mitchell      | 1 - Critical | On Hold     | Software       | Service Desk            | (empty)       | 2015-11-24 07:47:36 | admin      |
| INC0000553 | 2024-05-21 13:48:46 | The SAP HR application is not accessible                        | Margaret Grey          | 1 - Critical | In Progress | Inquiry / Help | Software                | Beth Anglin   | 2024-06-11 12:42:25 | admin      |
| INC0000552 | 2024-05-21 13:48:40 | SAP Financial Accounting application appears to be down         | Bud Richman            | 1 - Critical | In Progress | Software       | Software                | Fred Luddy    | 2024-06-11 12:40:45 | system     |
| INC0000551 | 2024-05-21 13:48:32 | Manager can't access SAP Controlling application                | Joe Employee           | 1 - Critical | In Progress | Software       | Software                | Don Goodliffe | 2024-06-11 12:40:45 | admin      |
| INC0000550 | 2024-05-21 14:58:24 | Can't access Exchange server - Is it down?                      | Jerrod Bennett         | 1 - Critical | In Progress | Hardware       | Hardware                | Beth Anglin   | 2024-06-11 12:49:40 | admin      |
| INC000049  | 2024-05-21 14:56:37 | Network storage unavailable                                     | Beth Anglin            | 2 - High     | In Progress | Network        | Hardware                | Don Goodliffe | 2024-06-11 12:52:34 | admin      |

Figure 11: Alert shown when State is edited from the list, confirming the change is blocked.

Figure 12: State changed to In Progress from the Incident form, which is allowed.

| Number     | Opened              | Short description                                               | Caller                 | Priority     | State       | Category       | Assignment group        | Assigned to  | Updated             | Updated by |
|------------|---------------------|-----------------------------------------------------------------|------------------------|--------------|-------------|----------------|-------------------------|--------------|---------------------|------------|
| INC0010002 | 2024-09-28 03:11:25 | Testing UI policy                                               | Adela Cervantoz        | 2 - High     | In Progress | Software       | (empty)                 | (empty)      | 2024-09-28 03:27:30 | admin      |
| INC0009009 | 2018-08-30 01:06:16 | Unable to access the shared folder.                             | David Miller           | 1 - Critical | New         | Inquiry / Help | Application Development | (empty)      | 2018-09-28 02:03:22 | system     |
| INC0009005 | 2018-09-31 21:55:21 | Email server is down.                                           | David Miller           | 1 - Critical | New         | Software       | (empty)                 | (empty)      | 2018-12-12 23:18:55 | admin      |
| INC0009001 | 2018-09-11 20:56:26 | Unable to post content on a Wiki page                           | David Miller           | 3 - Moderate | New         | Inquiry / Help | (empty)                 | (empty)      | 2018-12-12 23:32:42 | admin      |
| INC0008112 | 2019-07-29 11:48:43 | Assessment - ATF Assessor                                       | survey user            | 5 - Planning | New         | Inquiry / Help | (empty)                 | (empty)      | 2019-07-29 11:49:28 | admin      |
| INC0008111 | 2019-07-22 14:04:57 | ATF: Test1                                                      | System Administrator   | 4 - Low      | In Progress | Inquiry / Help | (empty)                 | (empty)      | 2024-09-28 03:32:38 | admin      |
| INC0008001 | 2021-01-15 13:04:14 | ATF:TEST2                                                       | survey user            | 5 - Planning | New         | Inquiry / Help | (empty)                 | (empty)      | 2021-01-21 15:31:42 | admin      |
| INC0007002 | 2018-10-16 22:47:51 | Need access to the common drive.                                | David Miller           | 4 - Low      | New         | Inquiry / Help | (empty)                 | (empty)      | 2018-12-12 23:28:49 | admin      |
| INC0007001 | 2018-10-16 22:47:10 | Employee payroll application server is down.                    | David Miller           | 1 - Critical | New         | Hardware       | Opspace                 | (empty)      | 2024-06-12 15:15:11 | system     |
| INC0001990 | 2020-06-07 09:02:25 | Unable to access the personal details section in payroll portal | Problem CoordinatorATF | 5 - Planning | On Hold     | Inquiry / Help | (empty)                 | (empty)      | 2020-06-07 09:04:34 | admin      |
| INC0000559 | 2016-08-10 09:14:29 | Unable to access team file share                                | Rick Berle             | 3 - Moderate | New         | Inquiry / Help | (empty)                 | (empty)      | 2016-08-10 09:14:29 | admin      |
| INC0000558 | 2016-08-10 09:37:45 | Performance problems with email                                 | Bow Ruggeri            | 5 - Planning | New         | Inquiry / Help | (empty)                 | (empty)      | 2016-08-10 09:37:45 | admin      |
| INC0000557 | 2016-08-10 09:14:59 | Performance problems with wifi                                  | Bertie Luby            | 5 - Planning | New         | Inquiry / Help | (empty)                 | (empty)      | 2016-08-10 09:14:59 | admin      |
| INC0000555 | 2024-05-21 21:47:23 | SAP Sales app is not accessible                                 | Carol Coughlin         | 1 - Critical | In Progress | Service Desk   | Service Desk            | Beth Anglin  | 2024-06-04 16:08:31 | admin      |
| INC0000554 | 2015-11-02 12:49:08 | SAP Materials Management is slow or there is an outage          | Christen Mitchell      | 1 - Critical | On Hold     | Software       | Service Desk            | (empty)      | 2015-11-24 07:47:36 | admin      |
| INC0000553 | 2024-05-21 13:48:46 | The SAP HR application is not accessible                        | Margaret Grey          | 1 - Critical | In Progress | Inquiry / Help | Software                | Beth Anglin  | 2024-06-11 12:40:25 | admin      |
| INC0000552 | 2024-05-21 13:48:40 | SAP Financial Accounting application appears to be down         | Bud Richman            | 1 - Critical | In Progress | Software       | Software                | Fred Luddy   | 2024-06-04 14:50:24 | system     |
| INC0000551 | 2024-05-21 13:48:32 | Manager can't access SAP Controlling application                | Joe Employee           | 1 - Critical | In Progress | Software       | Software                | Don Goodlife | 2024-06-11 12:40:45 | admin      |
| INC0000550 | 2024-05-21 14:58:24 | Can't access Exchange server - is it down?                      | Jerrold Bennett        | 1 - Critical | In Progress | Hardware       | Hardware                | Beth Anglin  | 2024-06-11 12:49:40 | admin      |
| INC0000449 | 2024-05-21 14:56:37 | Network storage unavailable                                     | Beth Anglin            | 2 - High     | In Progress | Network        | Hardware                | Don Goodlife | 2024-06-11 12:52:34 | admin      |

Figure 13: Incident list showing INC0008111 with the updated State of In Progress.

## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- **Cleaner data:** required information is captured before a record is saved, which supports accurate reporting and SLA tracking.
- **Immediate feedback:** users are told what is wrong at the moment they act, on the same screen.
- **Consistency:** Urgency follows Impact automatically and cannot be edited on high-impact incidents.
- **Adaptive behaviour:** Reverse if false means the form returns to normal when the condition no longer applies.
- **Protection of State:** quick edits from the list cannot bypass the form.
- **Lightweight and simple:** a small set of configurations and short scripts that are easy to implement and understand.

### Disadvantages

- **Client-side only:** these rules run in the browser, so records created by other routes, such as imports or integrations, are not covered by them.
- **Urgency is not reset:** the onChange script sets Urgency when Impact becomes High but does not change it back when Impact is lowered; the field simply becomes editable again.
- **No exceptions for roles:** the onCellEdit script blocks every list edit of State, including for users who might legitimately need it.
- **Late warning for Assigned to:** the requirement is enforced when the user submits, so it is not flagged as mandatory while filling in the form.
- **Hard-coded values:** the impact and urgency values and message texts are written into the scripts, so changes require editing the scripts.

## 9. CONCLUSION

The Implement Client Script & UI Policy (Incident) project shows how UI Policies and Client Scripts can work together to shape the way an Incident form behaves and to keep incorrect data from being submitted. By combining a UI Policy and its actions with onChange, onSubmit and onCellEdit scripts, we made fields such as Assignment group, Assigned to and Urgency respond sensibly to changes in Impact, which keeps the data clean and consistent.

Testing confirmed each rule in practice: the save was blocked with a clear message when Assigned to was missing, Urgency followed a High impact, the fields returned to normal when Impact was lowered, and State could not be changed from the list but could be changed from the form. The solution is lightweight, efficient and

easy to implement, which makes it well suited to a short micro project while still reflecting good practice for improving form usability and data integrity in ServiceNow.

## 10. FUTURE SCOPE

There are several natural ways to build on this work:

- Reset Urgency to a suitable value when Impact is lowered, so the two fields stay aligned in both directions.
- Add role-based exceptions to the onCellEdit script so authorised users can still make list edits where appropriate.
- Show Assigned to as mandatory on the form itself for High impact, so users see the requirement before they submit.
- Reinforce the rules with server-side controls so that records created through imports or integrations follow the same standards.
- Extend the approach to other impact levels, other fields and other tables, using the same pattern of a UI Policy plus supporting Client Scripts.
- Move message texts and values into configurable messages or properties so they can be changed without editing scripts.

## 11. APPENDIX

### Source Code

The client scripts below are those configured in the ServiceNow instance.

#### Client Script 1: Auto set urgency for high impact (onChange, field: Impact)

```
function onChange(control, oldValue, newValue, isLoading) {
  if (isLoading || newValue == '') {
    return;
  }
  if (newValue == '1') { // High Impact
    g_form.setValue('urgency', '1');
    g_form.showFieldMsg(
      'impact',
      'High impact incidents must have High urgency and an assignment group.',
      'info'
    );
  }
}
```

#### Client Script 2: Prevent save if assigned to missing (onSubmit)

```
function onSubmit() {
  if (g_form.getValue('impact') == '1' &&
    g_form.getValue('assigned_to') == '') {
    g_form.showErrorBox(
      'assigned_to',
      'Assigned To is mandatory for High impact incidents.'
    );
    return false;
  }
  return true;
}
```

**Client Script 3: Prevent state change via list edit (onCellEdit, field: State)**

```
function onCellEdit(sysIDs, table, oldValues, newValue, callback) {
  alert('State cannot be changed from the list. Please open the Incident.');
```

```
  // Cancel the list edit
  callback(false, oldValues);
}
```

**Dataset Link**

Not applicable. The project works on the standard Incident table in the ServiceNow instance, and no external dataset was used.

**GitHub & Project Demo Link**

| Item               | Link                    |
|--------------------|-------------------------|
| GitHub repository  | To be added by the team |
| Project demo video | To be added by the team |